# Chapter 6. Normed Spaces

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## Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter studies general normed spaces: dual spaces (of $\ell^p$ and of Hilbert spaces), and bounded linear maps between Banach spaces (Baire category, open mapping, closed graph, uniform boundedness), with applications to Fourier transforms.

## 6.1 Dual Spaces

**Motivation.** A normed space is understood through its continuous linear functionals; identifying duals, and having enough functionals to separate points (Hahn–Banach), is the basic tool.

### 6.1.1 Bounded linear maps

**Definition (operator norm)**

For a linear map $T : X \to Y$ between normed spaces put

$$\|T\| = \sup\{\|T(x)\| : x \in X,\ \|x\| \le 1\}.$$

If $\|T\| < \infty$, $T$ is a **bounded linear map** and $\|T\|$ is its **norm**.

The norm of $T$ measures how much $T$ stretches the unit ball $\{x \in X : \|x\| \le 1\}$ of $X$ inside $Y$. For example, $z \mapsto \alpha z : \mathbb{C} \to \mathbb{C}$ has norm $|\alpha|$.

**Exercise 6.1.1**

Show that a linear map $T : X \to Y$ is bounded if and only if there is $M > 0$ with

$$x \in X \implies \|T(x)\| \le M\|x\|,$$

and that in this case $\|T\| \le M$.

**Solution** [Supplement]. If $T$ is bounded, take $M = \|T\|$ (or $\|T\|+1$): $\|T(x)\| = \|x\|\,\|T(x/\|x\|)\| \le \|T\|\|x\|$ for $x \neq 0$. Conversely $\|T(x)\| \le M\|x\|$ gives $\|T\| \le M$. □

**Exercise 6.1.2**

Prove that $\|T\| = \sup\left\{\dfrac{\|T(x)\|}{\|x\|} : x \in X,\ x \neq 0\right\}$.

**Solution** [Supplement]. Let $S$ be the right side. For $x \neq 0$, $\|T(x)\|/\|x\| = \|T(x/\|x\|)\| \le \|T\|$, so $S \le \|T\|$. Conversely, if $0 < \|x\| \le 1$ then $\|T(x)\| \le \|T(x)\|/\|x\| \le S$, and $T(0) = 0$; so $\|T\| \le S$. □

**Exercise 6.1.3**

Show that the set of all bounded linear maps between two normed spaces is a vector space, and that $T \mapsto \|T\|$ is a norm on it.

**Solution** [Supplement]. If $S, T$ are bounded and $a \in \mathbb{C}$, then for $\|x\| \le 1$

$$\|(S+T)(x)\| \le \|S(x)\| + \|T(x)\| \le \|S\| + \|T\|, \qquad \|(aT)(x)\| = |a|\|T(x)\| \le |a|\|T\|,$$

so $S+T$, $aT$ are bounded with $\|S+T\| \le \|S\| + \|T\|$, $\|aT\| \le |a|\|T\|$, and equality in the latter via $a^{-1}$. $\|T\| = 0$ forces $T = 0$ on the unit ball, hence everywhere. □

**Proposition 6.1.1**

For a linear map $T : X \to Y$ between normed spaces, the following are equivalent:

(a) $T$ is bounded;

(b) $T$ is continuous;

(c) $T$ is continuous at some point $x_0 \in X$.

**Proof.** $\|T(x) - T(y)\| \le \|T\|\|x - y\|$ gives (a)⇒(b); (b)⇒(c) is trivial. If $T$ is continuous at $x_0$, $\|x - x_0\| < \delta \Rightarrow \|T(x) - T(x_0)\| < 1$, i.e. $\|y\| < \delta \Rightarrow \|T(y)\| < 1$, so $\|T\| \le 1/\delta$. ■

**Proposition 6.1.2**

For normed spaces $X, Y$, the set $B(X, Y)$ of bounded linear maps from $X$ to $Y$ is a normed space. If $Y$ is a Banach space, so is $B(X, Y)$.

**Proof.** $B(X, Y)$ is normed (Exercise 6.1.3). For $Y$ Banach and $\langle T_n \rangle$ Cauchy, $\|T_n(x) - T_m(x)\| \le \|T_n - T_m\|\|x\|$, so $T(x) = \lim T_n(x)$ exists, and $T \in B(X, Y)$ with $\|T_n - T\| \to 0$ (Exercise 6.1.4). ■

**Exercise 6.1.4**

Complete the proof of Proposition 6.1.2.

**Solution** [Supplement]. $T$ is linear. If $\|T_n - T_m\| < \epsilon$ for $m, n \ge N$, then $\|T_n(x) - T_m(x)\| < \epsilon$ for $\|x\| \le 1$, and $m \to \infty$ gives $\|T_n(x) - T(x)\| \le \epsilon$. So $T = T_N - (T_N - T) \in B(X, Y)$ and $\|T_n - T\| \le \epsilon$ for $n \ge N$. □

**Definition (dual space)**

A scalar-valued linear map is a **linear functional**; $X^* = B(X, \mathbb{C})$, the **dual space**, is a Banach space by Proposition 6.1.2.

**Example 1**

$f \mapsto \int_0^1 f$ is a bounded linear functional on $(C[0,1], \|\cdot\|_{\sup})$. However

$$f \mapsto \int_{\mathbb{R}} f(x)\, dx \tag{1}$$

is not even defined on $C_0(\mathbb{R})$ (integrals may diverge), and on $(C_c(\mathbb{R}), \|\cdot\|_{\sup})$ it is unbounded: $f_n \in C_c(\mathbb{R})$, $0 \le f_n \le 1$, $f_n = 1$ on $[-n, n]$ give $\|f_n\|_{\sup} = 1$, $\int f_n \to \infty$. On $(L^1(\mathbb{R}), \|\cdot\|_1)$ it is bounded by Fact B.9.

**Example 2**

Let $(X, \mu)$ be a measure space, $p, q$ conjugate exponents, and $g \in L^q(\mu)$ $(1 \le q \le \infty)$. Consider

$$\varphi_g : f \mapsto \int_X fg\, d\mu, \qquad f \in L^p(\mu).$$

By Hölder, $|\varphi_g(f)| \le \|g\|_q \|f\|_p$, so $\varphi_g$ is a bounded linear functional on $L^p(\mu)$ with

$$\|\varphi_g\| \le \|g\|_q, \qquad g \in L^q(\mu).$$

In fact equality holds.
*Case* $p = \infty$, $q = 1$. For fixed $g \in L^1(\mu)$ we want $f \in L^\infty(\mu)$ with $\|f\|_\infty \|g\|_1 = |\int_X gf|$. Take

$$f(x) = \begin{cases} \dfrac{|g(x)|}{g(x)}, & g(x) \neq 0, \\ 0, & g(x) = 0; \end{cases}$$

this function is written $\operatorname{sgn} g$. Then $gf = |g|$, so $\int_X gf = \|g\|_1 = \|f\|_\infty \|g\|_1$ (if $g \neq 0$).
*Case* $p = 1$, $q = \infty$. For $g \in L^\infty(\mu)$, $\|g\|_\infty \neq 0$, and small $\epsilon > 0$, $\{|g| > \|g\|_\infty - \epsilon\}$ has positive measure; take $E$ in it with $0 < \mu(E) < \infty$ ($\sigma$-finiteness) and $f = \chi_E \cdot \operatorname{sgn} g \in L^1(\mu)$. Then

$$\int_X gf = \int_E g \cdot \operatorname{sgn} g = \int_E |g| \ge \mu(E)(\|g\|_\infty - \epsilon) = \|f\|_1(\|g\|_\infty - \epsilon),$$

so $\|\varphi_g\| \ge \|g\|_\infty - \epsilon$, and hence $\|\varphi_g\| = \|g\|_\infty$.
*Case* $1 < q < \infty$. For fixed $g \in L^q(\mu)$ put $f = |g|^{q/p} \operatorname{sgn} g$. Then $|f|^p = |g|^q$, so $\|g\|_q^{q-1} = \|f\|_p$ (as $q/p = q - 1$), and $fg = |g|^q$. Hence

$$\int_X fg = \int |g|^q = \|g\|_q^q = \|g\|_q \|f\|_p,$$

and $\|\varphi_g\| = \|g\|_q$.

So for any measure space $(X, \mu)$ and $q \in [1, \infty]$,

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^* \tag{2}$$

is a norm-preserving linear map, an **isometry** (hence injective) (for $p = 1$, $q = \infty$ assume $\sigma$-finiteness).

**Exercise 6.1.5**

For the bounded function $g(x) = x$ on $[0, 1]$, show that there is no $f \in L^1[0, 1]$, $f \neq 0$, with $\|f\|_1 \|g\|_\infty = \int_0^1 fg$.

**Solution** [Supplement]**.** Here $\|g\|_\infty = 1$. For any $f \in L^1[0, 1]$,

$$\Big| \int_0^1 f(x) x \, dx \Big| \le \int_0^1 |f(x)| x \, dx \le \int_0^1 |f(x)| \, dx = \|f\|_1 \|g\|_\infty.$$

Equality would force $\int_0^1 |f(x)|(1 - x) \, dx = 0$, so $|f(x)|(1 - x) = 0$ a.e.; since $1 - x > 0$ on $[0, 1)$, $f = 0$ a.e. So the supremum $\|\varphi_g\| = 1$ is not attained. □

### 6.1.2 The duals of $\ell^p$ and $c_0$

In many cases (2) is surjective; we examine $\ell^p$. For $n = 1, 2, \ldots$ define the sequence

$$e_n(m) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{3}$$

If $\varphi \in (\ell^p)^*$ and $\varphi = \varphi_g$, then

$$\varphi(e_n) = \varphi_g(e_n) = \sum_{i=1}^{\infty} g(i) e_n(i) = g(n), \qquad n \in \mathbb{N}.$$

So for arbitrary $\varphi \in (\ell^p)^*$ define $g(n) = \varphi(e_n)$. Assume $1 \le p < \infty$; it suffices to show $g \in \ell^q$ (and $\varphi = \varphi_g$) to prove (2) is surjective.

$p = 1$, $q = \infty$. If $\varphi \in (\ell^1)^*$, then since $e_n \in \ell^1$, $|\varphi(e_n)| \le \|\varphi\|\|e_n\|_1 = \|\varphi\|$ for each $n$, so $g \in \ell^\infty$ and $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. Fix $N \in \mathbb{N}$ and, as in Example 2, put

$$f(n) = \begin{cases} |g(n)|^{q-1} \cdot \operatorname{sgn} g(n), & n \le N, \\ 0, & n > N. \end{cases}$$

Then $\|f\|_p^p = \sum_{n=1}^N |f(n)|^p = \sum_{n=1}^N |g(n)|^q$, so

$$\sum_{n=1}^N |g(n)|^q = \sum_{n=1}^\infty f(n)g(n) = \varphi(f) \le \|\varphi\|\|f\|_p = \|\varphi\|\Big(\sum_{n=1}^N |g(n)|^q\Big)^{1/p}.$$

Thus $\big(\sum_{n=1}^N |g(n)|^q\big)^{1/q} \le \|\varphi\|$ for every $N$, so $\|g\|_q \le \|\varphi\|$ and $g \in \ell^q$.

Finally $\varphi(e_n) = g(n) = \varphi_g(e_n)$, so $\varphi = \varphi_g$ on the finite combinations of the $e_n$, dense in $\ell^p$ ($p < \infty$); both are continuous, so $\varphi_g = \varphi$.

**Theorem 6.1.3**

If $1 \le p < \infty$ and $p, q$ are conjugate exponents, then $g \mapsto \varphi_g : \ell^q \to (\ell^p)^*$ is a surjective isometry.

By Hahn–Banach (below), $g \mapsto \varphi_g$ is not surjective for $p = \infty$: $(\ell^\infty)^* \supsetneq \ell^1$. Restricting functionals of $\ell^\infty$ to $c_0 = c_0(\mathbb{N})$, consider

$$g \mapsto \varphi_g : \ell^1 \to (c_0)^*. \tag{4}$$

**Exercise 6.1.6**

Show that (4) is a surjective isometry.

**Solution** [Supplement]. *Isometry.* $|\varphi_g(x)| = |\sum g(n)x(n)| \le \|g\|_1\|x\|_\infty$, so $\|\varphi_g\| \le \|g\|_1$. For $N \in \mathbb{N}$ let $x_N(n) = \operatorname{sgn} g(n)$ for $n \le N$ and 0 for $n > N$. Then $x_N \in c_0$, $\|x_N\|_\infty \le 1$, and $\varphi_g(x_N) = \sum_{n \le N} |g(n)|$. Letting $N \to \infty$ gives $\|\varphi_g\| \ge \|g\|_1$. *Surjectivity.* For $\varphi \in (c_0)^*$ put $g(n) = \varphi(e_n)$; with $x_N$ as above, $\sum_{n \le N} |g(n)| = \varphi(x_N) \le \|\varphi\|$, so $\|g\|_1 \le \|\varphi\|$. $\varphi$ and $\varphi_g$ agree on finite combinations of the $e_n$, dense in $c_0$ ($\|x - x\chi_{[1,N]}\|_\infty = \sup_{n>N}|x(n)| \to 0$), so $\varphi = \varphi_g$. □

Summarizing:

$$(c_0)^* = \ell^1, \qquad (\ell^p)^* = \ell^q \quad (1 \le p < \infty),$$

and therefore in many cases $X = X^{**}$.

### 6.1.3 The canonical embedding $X \to X^{**}$

For each $x \in X$ define

$$T_x(\varphi) = \varphi(x), \qquad \varphi \in X^*. \tag{5}$$

Clearly $T_x$ is a linear functional on $X^*$ and $\|T_x\| \le \|x\|$, so $T_x \in X^{**}$. To show that $x \mapsto T_x$ is injective we must show

$$x \ne 0 \implies T_x \ne 0,$$

i.e. for $x \ne 0$ some $\varphi \in X^*$ has $\varphi(x) \ne 0$: does $X$ have enough functionals to separate points? The next theorem says yes.

**Theorem 6.1.4 (Hahn–Banach)**

Let $X$ be a normed space and $Y$ a subspace. For every $\psi \in Y^*$ there is $\varphi \in X^*$ with

$$\varphi|_Y = \psi, \qquad \|\varphi\| = \|\psi\|.$$

**Proof.** Prove it first for real spaces: $X, Y$ real, $\psi$ real. If $\psi = 0$ take $\varphi = 0$; else normalize $\|\psi\| = 1$. Step 1 (the heart) adds one dimension; Step 2 iterates it via the axiom of choice; Step 3 passes to complex spaces.

*Step 1 (one more dimension).* For $x_0 \in X \setminus Y$ consider

$$Y_1 = \{ax_0 + y : a \in \mathbb{R},\ y \in Y\},$$

and extend $\psi$ to $\psi_1 : Y_1 \to \mathbb{R}$. Only $\psi_1(x_0)$ is to be chosen, so that $\|\psi_1\| = 1$, i.e.

$$|a\psi_1(x_0) + \psi(y)| \le \|ax_0 + y\|, \qquad a \in \mathbb{R},\ y \in Y. \tag{6}$$

For $y, y' \in Y$,

$$\psi(y) - \psi(y') = \psi(y - y') \le \|y - y'\| \le \|y - x_0\| + \|y' - x_0\|,$$

so

$$\psi(y) - \|y - x_0\| \le \psi(y') + \|y' - x_0\|, \qquad y, y' \in Y.$$

Hence with

$$A = \sup\{\psi(y) - \|y - x_0\| : y \in Y\}, \qquad B = \inf\{\psi(y') + \|y' - x_0\| : y' \in Y\},$$

we have $A \le B$. Choose any number in $[A, B]$ as $\psi_1(x_0)$. Then for all $y, y' \in Y$

$$\psi(y) - \|y - x_0\| \le \psi_1(x_0) \le \psi(y') + \|y' - x_0\|,$$

equivalent to $|\psi_1(x_0) - \psi(y)| \le \|x_0 - y\|$ for $y \in Y$; applied to $-\frac{y}{a}$ ($a \neq 0$) and multiplied by $|a|$, it gives (6) ($a = 0$ is trivial).

*Step 2 (Zorn).* Define

$$\mathcal{P} = \{(Y_\iota, \psi_\iota) :\ Y_\iota \text{ a subspace of } X \text{ containing } Y,\ \psi_\iota \in Y_\iota^*,\ \psi_\iota|_Y = \psi,\ \|\psi_\iota\| = 1\}.$$

Order $\mathcal{P}$ by $(Y_1, \psi_1) \le (Y_2, \psi_2)$ if $Y_1 \subset Y_2$ and $\psi_1 = \psi_2|_{Y_1}$. By the axiom of choice (Hausdorff maximality) there is a maximal totally ordered $\Omega \subset \mathcal{P}$ (no totally ordered $\mathcal{N} \supsetneq \Omega$). Then

$$Z = \bigcup\{Y_\iota : (Y_\iota, \psi_\iota) \in \Omega\}$$

is a subspace, and $\rho(x) = \psi_\iota(x)$ for $x \in Y_\iota$ defines a linear functional on $Z$ with $\|\rho\| = 1$ (Exercise 6.1.7). If $Z \subsetneq X$, Step 1 gives $(Z_1, \rho_1) \in \mathcal{P}$, $Z_1 \supsetneq Z$, and $\Omega \cup \{(Z_1, \rho_1)\}$ is totally ordered, contradicting maximality. So $Z = X$.

*Step 3 (complex case).* For complex $X, Y$, $\psi$, regard them as real spaces and write $\psi = u + iv$; $u$ is real linear on $Y$ with $\|u\| \le \|\psi\|$, and the real case gives $U$ on $X$ with $U|_Y = u$, $\|U\| = \|u\|$. For $x \in X$ define

$$\varphi(x) = U(x) - iU(ix).$$

Clearly $\varphi(x + y) = \varphi(x) + \varphi(y)$, and

$$\varphi(ix) = U(ix) - iU(-x) = i[U(x) - iU(ix)] = i\varphi(x),$$

so $\varphi$ is complex linear. For $y \in Y$,

$$\varphi(y) = U(y) - iU(iy) = u(y) - iu(iy),$$

so to show $\varphi|_Y = \psi$ it suffices that $\psi(y) = u(y) - iu(iy)$. For any $\alpha \in \mathbb{C}$ one checks immediately $\alpha = \operatorname{Re}\alpha - i\operatorname{Re}(i\alpha)$; since $\psi$ is complex linear,

$$\psi(y) = \operatorname{Re}\psi(y) - i\operatorname{Re}(i\psi(y)) = u(y) - i\operatorname{Re}(\psi(iy)) = u(y) - iu(iy),$$

so $\varphi|_Y = \psi$.

Finally $\|\varphi\| = \|\psi\|$: $\|\varphi\| \geq \|\psi\|$ as $\varphi$ extends $\psi$, and it suffices that $\|\varphi\| \leq \|U\| \leq \|\psi\|$. For $x \in X$ take $|\alpha| = 1$ with $\alpha\varphi(x) = |\varphi(x)|$; then $\varphi(\alpha x)$ is real, so

$$|\varphi(x)| = \varphi(\alpha x) = U(\alpha x) \leq \|U\|\|\alpha x\| = \|U\|\|x\|.$$

Thus $\|\varphi\| \leq \|U\|$, completing the proof. ∎

**Exercise 6.1.7**

In the proof of the Hahn–Banach theorem, show that $Z$ is a subspace and that $\rho$ is a linear functional on $Z$ with $\|\rho\| = 1$.

**Solution** [Supplement]. *$Z$ is a subspace.* Let $x, x' \in Z$, say $x \in Y_\iota$, $x' \in Y_\kappa$ with $(Y_\iota, \psi_\iota), (Y_\kappa, \psi_\kappa) \in \Omega$. Since $\Omega$ is totally ordered, one contains the other, say $Y_\iota \subset Y_\kappa$; then $ax + bx' \in Y_\kappa \subset Z$. *$\rho$ is well defined.* If $x \in Y_\iota \cap Y_\kappa$ and $Y_\iota \subset Y_\kappa$, then $\psi_\iota = \psi_\kappa|_{Y_\iota}$, so $\psi_\iota(x) = \psi_\kappa(x)$. *Linearity.* With $x, x' \in Y_\kappa$ as above, $\rho(ax + bx') = \psi_\kappa(ax + bx') = a\psi_\kappa(x) + b\psi_\kappa(x') = a\rho(x) + b\rho(x')$. *Norm.* For $x \in Y_\iota$, $|\rho(x)| = |\psi_\iota(x)| \leq \|x\|$, so $\|\rho\| \leq 1$; and $\rho|_Y = \psi$ with $\|\psi\| = 1$ gives $\|\rho\| \geq 1$. □

For $x \in X$, $x \neq 0$, apply Hahn–Banach to the one-dimensional subspace $Y = \{\alpha x : \alpha \in \mathbb{C}\}$ and the functional $\psi(\alpha x) = \alpha\|x\|$ (which has $\|\psi\| = 1$):

**Corollary 6.1.5**

If $x \neq 0$ is a point of a normed space $X$, there is $\varphi \in X^*$ with $\varphi(x) = \|x\|$ and $\|\varphi\| = 1$.

By Corollary 6.1.5, $\|T_x\| = \|x\|$ in (5): $x \mapsto T_x$ embeds $X$ isometrically in $X^{**}$. If it is surjective, $X$ is **reflexive**; e.g. $\ell^p$ $(1 < p < \infty)$, but not $c_0$, as $(c_0)^{**} = \ell^\infty$.

The next corollary is useful for deciding whether a given subspace is closed. The closure of a subset $Y$ is written $\overline{Y}$.

**Corollary 6.1.6**

For a subspace $Y$ of a normed space $X$ and a point $x \in X$, the following are equivalent:

(a) $x \in \overline{Y}$;

(b) if $\varphi \in X^*$ and $\varphi|_Y = 0$, then $\varphi(x) = 0$.

**Proof.** Every $\varphi \in X^*$ is continuous, so (a)⇒(b) is clear. If $x \notin \overline{Y}$, there is $\delta > 0$ with

$$y \in Y \implies \|x - y\| > \delta.$$

Consider the subspace $Z = \{\alpha x + y : \alpha \in \mathbb{C},\ y \in Y\}$ spanned by $x$ and $Y$, and define $\psi(\alpha x + y) = \alpha$ (well defined since $x \notin Y$); $\psi$ is a linear functional on $Z$. For $\alpha x + y \in Z$ with $\alpha \neq 0$,

$$|\psi(\alpha x + y)|\delta = |\alpha|\delta \leq |\alpha|\left\|x + \frac{1}{\alpha}y\right\| = \|\alpha x + y\|,$$

so $\psi \in Z^*$ with $\|\psi\| \leq \frac{1}{\delta}$. Also $\psi(x) = 1 \neq 0$. Extending $\psi$ to $X$ by Hahn–Banach gives $\varphi \in X^*$ with $\varphi|_Y = 0$ but $\varphi(x) \neq 0$. ∎

**Exercise 6.1.8**

Show that there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ but $\varphi \neq 0$. Show that $(\ell^\infty)^* \supsetneq \ell^1$, and that the Banach space $\ell^1$ is not reflexive.

**Solution** [Supplement]. $c_0$ is a closed subspace of $\ell^\infty$ (Exercise 5.1.9) and the constant sequence $\mathbf{1} \notin c_0$, so $\mathbf{1} \notin \overline{c_0}$. By Corollary 6.1.6 there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ and $\varphi(\mathbf{1}) \neq 0$. Such $\varphi$ is not of the form $\varphi_g$, $g \in \ell^1$: otherwise $g(n) = \varphi_g(e_n) = \varphi(e_n) = 0$ for all $n$ (as $e_n \in c_0$), so $g = 0$ and $\varphi = 0$. Hence the isometry $g \mapsto \varphi_g : \ell^1 \to (\ell^\infty)^*$ is not surjective: $(\ell^\infty)^* \supsetneq \ell^1$. $\ell^1$ not reflexive: $(\ell^1)^* = \ell^\infty$ (Theorem 6.1.3), so $(\ell^1)^{**} = (\ell^\infty)^*$. If $x \mapsto T_x$ were onto, then $\varphi = T_g$ for some $g \in \ell^1$, i.e. $\varphi(h) = T_g(\varphi_h) = \sum_n g(n)h(n)$ for $h \in \ell^\infty$ (identifying $h$ with $\varphi_h \in (\ell^1)^*$); so $\varphi = \varphi_g$, which we just ruled out. □

**Exercise 6.1.9**

Show that a Banach space $X$ is reflexive if and only if $X^*$ is reflexive. Show that the Banach space $\ell^\infty$ is not reflexive.

**Solution** [Supplement]. Write $J_X : X \to X^{**}$ and $J_{X^*} : X^* \to X^{***}$ for the canonical isometries.

($\Rightarrow$) Let $X$ be reflexive and $\Phi \in X^{***}$. Put $\varphi = \Phi \circ J_X \in X^*$. For $\xi \in X^{**}$ write $\xi = J_X(x)$; then $\Phi(\xi) = \Phi(J_X x) = \varphi(x) = J_X(x)(\varphi) = \xi(\varphi) = J_{X^*}(\varphi)(\xi)$. So $\Phi = J_{X^*}(\varphi)$ and $X^*$ is reflexive.

($\Leftarrow$) If $X^*$ is reflexive and $J_X(X) \subsetneq X^{**}$, then $J_X(X)$ is closed ($X$ complete), so Corollary 6.1.6 gives $0 \neq \Phi \in X^{***}$ vanishing on it, $\Phi = J_{X^*}(\varphi)$. Then $\varphi(x) = \Phi(J_X x) = 0$ for all $x$, so $\varphi = 0$ and $\Phi = 0$: a contradiction. Finally $\ell^\infty = (\ell^1)^*$ and $\ell^1$ is not reflexive (Exercise 6.1.8), so $\ell^\infty$ is not reflexive. □

## 6.2 Linear Maps between Banach Spaces

**Motivation.** Via Baire's theorem, completeness makes bijective bounded maps invertible, closed graphs bounded, and pointwise bounded families uniformly bounded; one application is a continuous function with divergent Fourier series.

Completeness now matters: the domain or range is a Banach space. The key property of complete metric spaces is Baire's theorem. For $x$ in a metric space $(X, d)$ and $r > 0$ write

$$N_X(x, r) = \{y \in X : d(x, y) < r\}$$

for the neighborhood of $x$, or just $N(x, r)$. When $X$ is a normed space, $N_X(0, r)$ is written $X_r$.

**Theorem 6.2.1 (Baire)**

If a sequence $\langle C_n \rangle$ of closed subsets of a complete metric space $X$ has union $X$, then some $C_n$ contains a nonempty open set.

**Proof.** Suppose not. As $C_1 \neq X$ is closed, some $N(x_1, r_1)$ has $C_1 \cap \overline{N(x_1, r_1)} = \emptyset$ ($\overline{A}$ the closure). As $N(x_1, r_1) \not\subset C_2$, there is $N(x_2, r_2)$ with

$$\overline{N(x_2, r_2)} \subset N(x_1, r_1) \setminus C_2$$

($N(x_1, r_1) \setminus C_2$ is a nonempty open set). Repeating, we get neighborhoods $\langle N(x_n, r_n) \rangle$ with

$$N(x_{n+1}, r_{n+1}) \subset N(x_n, r_n), \qquad \overline{N(x_n, r_n)} \cap \Big[ \bigcup_{k=1}^{n} C_k \Big] = \emptyset$$

for each $n$. We may also arrange $r_n \to 0$; then $\langle x_n \rangle$ is a Cauchy sequence (for $m > n$, $x_m \in N(x_n, r_n)$). Its limit $x$ lies in every $\overline{N(x_n, r_n)}$, so $x \notin \bigcup_{n=1}^{\infty} C_n = X$, a contradiction. ■

**Corollary 6.2.2**

In a complete metric space, the intersection of a sequence of dense open sets is nonempty.

**Exercise 6.2.1**

Prove Corollary 6.2.2.

**Solution** [Supplement]. $C_n = U_n^c$ is closed with no interior. If $\bigcap_n U_n = \emptyset$, then $\bigcup_n C_n = X$ and Baire gives some $C_n$ with interior: a contradiction. (Applied to $\overline{N(x,r)}$, this shows $\bigcap U_n$ is dense.) □

The open mapping theorem and uniform boundedness principle, with Hahn–Banach, are the three basic tools for Banach spaces.

### 6.2.1 The open mapping theorem

**Theorem 6.2.3 (open mapping theorem)**

Let $T : X \to Y$ be a surjective bounded linear map between Banach spaces. Then there is $\delta > 0$ with $T(X_1) \supset Y_\delta$.

**Proof.** First, $X = \bigcup_n X_n$ and $T$ is surjective, so $Y = \bigcup_n \overline{T(X_n)}$. By Baire's theorem there are $n \in \mathbb{N}$ and a neighborhood $N_Y(y_0, r)$ with $N_Y(y_0, r) \subset \overline{T(X_n)}$. If $\|y\| < r$, then $y + y_0 \in N_Y(y_0, r)$, so $y + y_0 \in \overline{T(X_n)}$. Choosing $x_0$ with $T(x_0) = y_0$,

$$y \in \overline{T(X_n)} - T(x_0) \subset \overline{T(X_{n+\|x_0\|})}.$$

Putting $\delta = \dfrac{r}{n + \|x_0\|}$ we get $Y_\delta \subset \overline{T(X_1)}$, and by homogeneity

$$Y_s \subset \overline{T(X_{s/\delta})}, \qquad s \in (0, \infty). \tag{7}$$

By (7) we show $Y_\delta \subset T(X_1)$ up to a factor. Fix $\epsilon \in (0,1)$. For $y \in Y_1$, (7) gives $x_1 \in X_{1/\delta}$ with $\|y - T(x_1)\| < \epsilon$; as $y - T(x_1) \in Y_\epsilon$, (7) with $s = \epsilon$ gives $x_2 \in X_{\epsilon/\delta}$ with $\|y - T(x_1) - T(x_2)\| < \epsilon^2$. Continuing,

$$\|x_n\| < \frac{\epsilon^{n-1}}{\delta}, \qquad \|y - T(x_1) - T(x_2) - \cdots - T(x_n)\| < \epsilon^n.$$

Hence $\sum_{n=1}^{\infty} x_n$ converges (absolutely, in the Banach space $X$) to some $x \in X$ with $\|x\| \le \frac{1}{\delta(1-\epsilon)}$, and

$$y = \lim_{n\to\infty} T(x_1 + x_2 + \cdots + x_n) = T(x).$$

So $y \in T(X_{1/\delta(1-\epsilon)})$; thus $Y_1 \subset T(X_{1/\delta(1-\epsilon)})$, i.e. $Y_{\delta(1-\epsilon)} \subset T(X_1)$. ■

$f : X \to Y$ is an **open map** if $f(U)$ is open for every open $U \subset X$; hence the name:

**Corollary 6.2.4**

A surjective bounded linear map $T : X \to Y$ between Banach spaces is always an open map.

**Exercise 6.2.2**

Prove Corollary 6.2.4.

**Solution** [Supplement]. Let $U \subset X$ be open and $y = T(x) \in T(U)$ with $x \in U$. Choose $r > 0$ with $N_X(x, r) = x + X_r \subset U$. By Theorem 6.2.3, $T(X_r) = rT(X_1) \supset rY_\delta = Y_{r\delta}$. Hence

$$N_Y(y, r\delta) = y + Y_{r\delta} \subset T(x) + T(X_r) = T(x + X_r) \subset T(U),$$

so $T(U)$ is open. □

**Corollary 6.2.5**

If a bounded linear map $T : X \to Y$ between Banach spaces is bijective, then its inverse $T^{-1} : Y \to X$ is also bounded.

**Proof.** Choose $\delta > 0$ with $Y_\delta \subset T(X_1)$. Then $T^{-1}(Y_\delta) \subset X_1$, so $\|T^{-1}\| \le \frac{1}{\delta}$. ■

### 6.2.2 The Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not onto

For $n = 1, 2, \dots$ let

$$D_n(t) = \sum_{k=-n}^{n} u_k(t) = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t}, \qquad t \in \mathbb{T}. \tag{8}$$

$\langle D_n \rangle$ is the **Dirichlet kernel**.

**Exercise 6.2.3**

Prove the second equality in (8).

**Solution** [Supplement]. For $t \neq 0 \pmod{2\pi}$, the geometric sum gives

$$\sum_{k=-n}^{n} e^{ikt} = e^{-int} \frac{e^{i(2n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t} - e^{-int}}{e^{it} - 1} = \frac{e^{i(n+\frac{1}{2})t} - e^{-i(n+\frac{1}{2})t}}{e^{it/2} - e^{-it/2}} = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t},$$

after multiplying numerator and denominator by $e^{-it/2}$. At $t = 0$ both sides equal $2n + 1$ (as a limit). □

Then $\widehat{D_n} = \sum_{k=-n}^{n} \chi_{\{k\}}$, so $\|\widehat{D_n}\|_\infty = 1$. On the other hand, since $|\sin t| \le |t|$,

$$\begin{aligned} \|D_n\|_1 &\ge \frac{1}{2\pi} \int_{-\pi}^{\pi} \Big| \frac{\sin(n + \frac{1}{2})t}{\frac{t}{2}} \Big| dt = \frac{2}{\pi} \int_0^{\pi} \frac{1}{t} \Big| \sin\Big(n + \frac{1}{2}\Big)t \Big| dt = \frac{2}{\pi} \int_0^{(n+\frac{1}{2})\pi} \frac{1}{t} |\sin t| \, dt \\ &\ge \frac{2}{\pi} \sum_{k=1}^{n} \frac{1}{k\pi} \int_{(k-1)\pi}^{k\pi} |\sin t| \, dt = \frac{4}{\pi^2} \sum_{k=1}^{n} \frac{1}{k}, \end{aligned}$$

so $\|D_n\|_1 \to \infty$, and $E_n = D_n / \|D_n\|_1$ has $\|E_n\|_1 = 1$, $\|\widehat{E_n}\|_\infty \to 0$. So no $\delta > 0$ has $\|\widehat{f}\|_\infty \ge \delta \|f\|_1$, as Corollary 6.2.5 would give for the injective $f \mapsto \widehat{f}$: *the Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not surjective.*

### 6.2.3 The closed graph theorem

The closed graph theorem, from the open mapping theorem, proves boundedness; the hypothesis says the graph of $T$ is closed.

**Theorem 6.2.6 (closed graph theorem)**

A linear map $T : X \to Y$ between Banach spaces satisfying

$$x_n \in X,\ \lim_{n\to\infty} x_n = x,\ \lim_{n\to\infty} T(x_n) = y \implies T(x) = y$$

is bounded.

**Proof.** $\|x\|' = \|x\| + \|T(x)\|$ is a norm on $X$. If $\langle x_n\rangle$ is $\|\cdot\|'$-Cauchy, then $\langle x_n\rangle$ and $\langle T(x_n)\rangle$ are Cauchy in the Banach spaces $X$, $Y$, so there are $x$, $y$ with

$$\lim_{n\to\infty}\|x_n - x\| = 0, \qquad \lim_{n\to\infty}\|T(x_n) - y\| = 0.$$

By hypothesis $T(x) = y$, so

$$\lim_{n\to\infty}\|x_n - x\|' = \lim_{n\to\infty}\|x_n - x\| + \lim_{n\to\infty}\|T(x_n) - T(x)\| = 0,$$

so $(X, \|\cdot\|')$ is Banach. As $\|x\| \le \|x\|'$, the identity $(X, \|\cdot\|') \to (X, \|\cdot\|)$ is a bounded bijection, with bounded inverse by Corollary 6.2.5: $\|x\|' \le M\|x\|$, and

$$\|Tx\| + \|x\| = \|x\|' \le M\|x\|, \qquad x \in X,$$

so $\|Tx\| \le (M-1)\|x\|$. ■

**Exercise 6.2.4**

In the proof of the closed graph theorem, show that $x \mapsto \|x\|'$ is a norm.

**Solution** [Supplement]. $\|x\|' \ge 0$, and $\|x\|' = 0$ forces $\|x\| = 0$, i.e. $x = 0$. By linearity of $T$, $\|ax\|' = |a|\|x\| + |a|\|Tx\| = |a|\|x\|'$ and

$$\|x + z\|' = \|x + z\| + \|Tx + Tz\| \le \|x\| + \|z\| + \|Tx\| + \|Tz\| = \|x\|' + \|z\|'.$$

□

### 6.2.4 The uniform boundedness principle

Another application of Baire's theorem is the uniform boundedness principle, also called the Banach–Steinhaus theorem.

**Theorem 6.2.7 (uniform boundedness principle)**

Let $\{T_\alpha : \alpha \in A\}$ be a family of bounded linear maps from a Banach space $X$ to a normed space $Y$. If for every $x \in X$

$$M(x) = \sup\{\|T_\alpha(x)\| : \alpha \in A\} < \infty,$$

then $\sup\{\|T_\alpha\| : \alpha \in A\} < \infty$.

**Proof.** For $n = 1, 2, \ldots$ put

$$C_n = \bigcap_{\alpha\in A}\{x \in X : \|T_\alpha(x)\| \le n\}.$$

Each $x \mapsto \|T_\alpha(x)\|$ is continuous, so $C_n$ is closed, and by hypothesis $X = \bigcup_n C_n$. By Baire's theorem some $C_n$ contains a neighborhood $N(x_0, r)$, i.e.

$$\alpha \in A,\ \|x - x_0\| < r \implies \|T_\alpha(x)\| \le n.$$

If $\|x\| < r$, then $\|(x + x_0) - x_0\| < r$, so for every $\alpha \in A$

$$\|T_\alpha(x)\| - \|T_\alpha(x_0)\| \le \|T_\alpha(x + x_0)\| \le n,$$

hence $\|T_\alpha(x)\| \le n + \|T_\alpha(x_0)\| \le n + M(x_0)$. Therefore

$$\sup_{\alpha \in A} \|T_\alpha\| \le \frac{1}{r}(n + M(x_0)).$$

■

The next corollary is very useful for defining new bounded linear maps from a given sequence.

**Corollary 6.2.8**

Let $T_n : X \to Y$ be bounded linear maps from a Banach space to a normed space with $\langle T_n(x) \rangle$ convergent for every $x$. Then $T(x) = \lim_n T_n(x)$ is a bounded linear map.

**Proof.** $T$ is linear, and convergent sequences are bounded, so uniform boundedness gives $M = \sup_n \|T_n\| < \infty$ and

$$\|T(x)\| = \lim_{n \to \infty} \|T_n(x)\| \le \sup_n \|T_n\| \|x\| = M\|x\|,$$

so $\|T\| \le M$. ■

### 6.2.5 A continuous function whose Fourier series diverges

By uniform boundedness, Fourier series of continuous functions need not converge pointwise. By (2) of §5.1 the partial sums are

$$\sum_{k=-n}^{n} \widehat{f}(k) u_k = \sum_{k=-n}^{n} u_k * f = D_n * f.$$

Fix $t_0 \in \mathbb{T}$ and $n$, and consider the linear functional on the Banach space $C(\mathbb{T})$

$$T_n : f \mapsto (D_n * f)(t_0), \qquad f \in C(\mathbb{T}).$$

Then

$$|T_n(f)| \le \int_{\mathbb{T}} |f(t) D_n(t_0 - t)| \, dt \le \|f\|_\infty \|D_n\|_1, \qquad f \in C(\mathbb{T}),$$

so $\|T_n\| \le \|D_n\|_1$. $\operatorname{sgn} D_n(t_0 - t)$ has finitely many discontinuities, so continuous $h_i$ with $\|h_i\|_\infty \le 1$ converge to it except at finitely many points, and by dominated convergence

$$T_n(h_i) = \int_{\mathbb{T}} h_i(t) D_n(t_0 - t) \, dt \to \int_{\mathbb{T}} |D_n(t_0 - t)| \, dt = \|D_n\|_1,$$

so $\|T_n\| = \|D_n\|_1 \to \infty$. If every $f \in C(\mathbb{T})$ had Fourier series converging at $t_0$, uniform boundedness would bound $\|T_n\|$. So *some continuous function has a Fourier series diverging at* $t_0$.

## 6.3 Hilbert Spaces

**Motivation.** Inner products bring Euclidean geometry to infinite dimensions: functionals are inner products, vectors expand in orthonormal bases, and $L^2(\mathbb{T})$ gives Fourier series.

### 6.3.1 Inner products

**Definition (inner product)**

For a complex vector space $X$, a function $(x,y)\mapsto\langle x,y\rangle : X\times X\to\mathbb{C}$ satisfying

(I1) $x\mapsto\langle x,y\rangle : X\to\mathbb{C}$ is a linear functional for each $y\in X$;

(I2) $\langle x,y\rangle=\overline{\langle y,x\rangle}$ for all $x,y\in X$;

(I3) $\langle x,x\rangle\geq 0$ for all $x\in X$;

(I4) $x=0 \iff \langle x,x\rangle=0$

is an **inner product**, and a vector space with an inner product is an **inner product space**.

**Example 1**

On $L^2(X,\mu)$,

$$\langle f,g\rangle=\int_X f\overline{g}\,d\mu,\qquad f,g\in L^2(X,\mu),$$

is an inner product.

For $x,y$ in an inner product space $X$, choose $\alpha\in\mathbb{C}$ with $|\alpha|=1$ and $\alpha\langle y,x\rangle=|\langle x,y\rangle|$. Then for every real $t$,

$$0\leq\langle x-t\alpha y,x-t\alpha y\rangle=\langle x,x\rangle-t\alpha\langle y,x\rangle-t\overline{\alpha}\langle x,y\rangle+t^2\langle y,y\rangle=\langle x,x\rangle-2t|\langle x,y\rangle|+t^2\langle y,y\rangle.$$

A nonnegative real quadratic in $t$ has nonpositive discriminant, so

$$|\langle x,y\rangle|^2\leq\langle x,x\rangle\langle y,y\rangle,\qquad x,y\in X.$$

This is the **Schwarz inequality**; for $L^2(X,\mu)$ it is a special case of Hölder's inequality. For $x\in X$ put

$$\|x\|=\langle x,x\rangle^{1/2};\tag{9}$$

$x\mapsto\|x\|$ is a norm, the triangle inequality following from Schwarz: $\|x+y\|^2=\|x\|^2+2\,\mathrm{Re}\langle x,y\rangle+\|y\|^2\leq(\|x\|+\|y\|)^2$. A complete inner product space is a **Hilbert space**. The norm determines the inner product by the **polarization identity**

$$\langle x,y\rangle=\frac{1}{4}\sum_{k=0}^{3}i^k\|x+i^ky\|^2,\tag{10}$$

and the norm of an inner product space satisfies the **parallelogram law**

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2.\tag{11}$$

**Exercise 6.3.1**

Prove (10) and (11) for vectors $x,y$ of an inner product space $X$.

**Solution** [Supplement]**.** For $c\in\mathbb{C}$ with $|c|=1$,

$$\|x+cy\|^2=\|x\|^2+\overline{c}\langle x,y\rangle+c\langle y,x\rangle+\|y\|^2.$$

(11): add the cases $c=1$ and $c=-1$; the cross terms cancel. (10): multiply the case $c=i^k$ by $i^k$ and sum over $k=0,1,2,3$:

$$\sum_k i^k\|x+i^ky\|^2=\Big(\sum_k i^k\Big)(\|x\|^2+\|y\|^2)+\Big(\sum_k i^k\overline{i^k}\Big)\langle x,y\rangle+\Big(\sum_k i^{2k}\Big)\langle y,x\rangle=0+4\langle x,y\rangle+0,$$

since $\sum_{k=0}^{3}i^k=0$, $i^k\overline{i^k}=1$, and $\sum_k(-1)^k=0$. □

### 6.3.2 The dual of a Hilbert space

We now determine the dual of a Hilbert space $H$. By the Schwarz inequality, for each $y \in H$

$$\phi_y : x \mapsto \langle x, y\rangle, \qquad x \in H,$$

is a bounded linear functional with $\|\phi_y\| = \|y\|$ (take $x = y$ for equality). Hence

$$y \mapsto \phi_y : H \to H^* \tag{12}$$

is an isometric injection, not linear but **conjugate linear**: $\phi_{\alpha y} = \overline{\alpha}\phi_y$.

To show (12) is onto, for $\phi \in H^*$ we need $y$ with $\phi(x) = \langle x, y\rangle$, i.e. a vector perpendicular to $\ker\phi$; the next theorem supplies one.

**Theorem 6.3.1 (closest point)**

Let $C$ be a closed convex subset of a Hilbert space $H$ and $x_0 \in H \setminus C$. Put

$$d = \inf\{\|x_0 - y\| : y \in C\}.$$

Then there is a unique $y_0 \in C$ with $\|x_0 - y_0\| = d$.

**Proof.** We may assume $x_0 = 0$ (translate $C$ by $-x_0$). Replacing $x, y$ by $\frac{1}{2}x, \frac{1}{2}y$ in the parallelogram law gives

$$\frac{1}{4}\|x - y\|^2 = \frac{1}{2}\|x\|^2 + \frac{1}{2}\|y\|^2 - \left\|\frac{x+y}{2}\right\|^2.$$

If $x, y \in C$ then $\frac{x+y}{2} \in C$ by convexity, so $\|\frac{x+y}{2}\| \geq d$ and

$$\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2 - 4d^2, \qquad x, y \in C. \tag{13}$$

*Uniqueness.* If $x, y \in C$ and $\|x\| = \|y\| = d$, (13) gives $x = y$. *Existence.* Take a sequence $\langle y_n\rangle$ in $C$ with $\|y_n\| \to d$. By (13),

$$\|y_n - y_m\|^2 \leq 2\|y_n\|^2 + 2\|y_m\|^2 - 4d^2 \to 0,$$

so $\langle y_n\rangle$ is Cauchy. As $H$ is complete and $C$ closed, $y_0 = \lim_n y_n \in C$, and $\|y_0\| = \lim_n \|y_n\| = d$. ■

Now let $E$ be a closed subspace of $H$ and $x_0 \in H \setminus E$. By Theorem 6.3.1 there is $y_0 \in E$ with

$$y \in E \implies \|x_0 - y_0\| \leq \|x_0 - y\|.$$

Put $z_0 = x_0 - y_0$. For every $y \in E$ and $\alpha \in \mathbb{C}$,

$$\langle z_0, z_0\rangle = \|x_0 - y_0\|^2 \leq \|x_0 - y_0 - \alpha y\|^2 = \langle z_0 - \alpha y, z_0 - \alpha y\rangle,$$

so

$$0 \leq -\alpha\langle y, z_0\rangle - \overline{\alpha}\langle z_0, y\rangle + |\alpha|^2\langle y, y\rangle.$$

Substituting $\alpha = t\langle z_0, y\rangle$ $(t \in \mathbb{R})$,

$$0 \leq -2t|\langle y, z_0\rangle|^2 + t^2|\langle y, z_0\rangle|^2\langle y, y\rangle$$

for every $t \in \mathbb{R}$; small $t > 0$ forces $\langle y, z_0\rangle = 0$ for $y \in E$. $x, y$ are **orthogonal**, $x \perp y$, if $\langle x, y\rangle = 0$; $E^\perp$ is the set of vectors orthogonal to $E$. As $z_0 \neq 0$ $(x_0 \notin E)$ and $z_0 \in E^\perp$:

**Corollary 6.3.2**

If $E$ is a closed proper subspace of a Hilbert space $H$, then $E^\perp \supsetneq \{0\}$.

Return to $\phi \in H^*$. If $\phi \neq 0$, apply Corollary 6.3.2 to the closed proper subspace $\ker\phi$ to get $z \in (\ker\phi)^\perp$ with $\|z\| = 1$. Since $\phi(x)z - \phi(z)x \in \ker\phi$, for every $x \in H$

$$0 = \langle \phi(x)z - \phi(z)x, z\rangle = \phi(x) - \phi(z)\langle x, z\rangle = \phi(x) - \langle x, \overline{\phi(z)}z\rangle.$$

So with $y = \overline{\phi(z)}z$,

$$\phi(x) = \langle x, y\rangle, \qquad x \in H.$$

This proves:

**Theorem 6.3.3 (Riesz representation)**

For a Hilbert space $H$, the map $y \mapsto \phi_y$ defined by (12) is a surjective, isometric, conjugate linear map $H \to H^*$.

**Exercise 6.3.2**

Show that every Hilbert space is reflexive.

**Solution** [Supplement]**.** By Theorem 6.3.3, $H^*$ is a Hilbert space with $\langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle$ (inducing the dual norm, as $y \mapsto \phi_y$ is a conjugate-linear isometric bijection). For $\Lambda \in H^{**}$, Theorem 6.3.3 on $H^*$ gives $\phi_z$ with $\Lambda(\phi) = \langle \phi, \phi_z\rangle_{H^*}$. Then for every $y$,

$$\Lambda(\phi_y) = \langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle = \phi_y(z) = T_z(\phi_y).$$

Since every element of $H^*$ is some $\phi_y$, $\Lambda = T_z$. So $x \mapsto T_x$ is onto. □

**Exercise 6.3.3**

Show that $E^\perp$ is a closed subspace for any subset $E$ of a Hilbert space. Show that $\overline{E} = E^{\perp\perp}$ for any subspace $E$.

**Solution** [Supplement]**.** $E^\perp = \bigcap_{y\in E} \ker\phi_y$ is an intersection of kernels of bounded linear functionals, hence a closed subspace. $E \subset E^{\perp\perp}$, closed, so $\overline{E} \subset E^{\perp\perp}$. Conversely, for $x \in E^{\perp\perp} \setminus \overline{E}$ the closest-point argument gives $y_0 \in \overline{E}$ with $z_0 = x - y_0 \neq 0$, $z_0 \in E^\perp$; also $z_0 \in E^{\perp\perp}$ (as $x, y_0 \in E^{\perp\perp}$), so $z_0 \perp z_0$, $z_0 = 0$: a contradiction. So $E^{\perp\perp} = \overline{E}$. □

### 6.3.3 Orthonormal sets

A subset $\{u_\alpha : \alpha \in A\}$ of an inner product space $H$ satisfying

$$\langle u_\alpha, u_\beta\rangle = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta \end{cases}$$

is an **orthonormal set**; the subspace it spans (finite linear combinations) is written $P_A$.

**Proposition 6.3.4**

Let $\{u_\alpha : \alpha \in F\}$ be a finite subset of an orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space $H$. For fixed $x \in H$,

$$\Big\|x - \sum_{\alpha\in F} \langle x, u_\alpha\rangle u_\alpha\Big\| \le \|x - y\|, \qquad y \in P_F, \tag{14}$$

with equality only for $y = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. Moreover

$$\sum_{\alpha\in F} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

**Proof.** Write $z = \sum_{\alpha\in F}\langle x, u_\alpha\rangle u_\alpha$. For each $\alpha \in F$, $\langle x - z, u_\alpha\rangle = \langle x, u_\alpha\rangle - \langle x, u_\alpha\rangle = 0$. If $y \in P_F$ then $y - z \in P_F$, so $\langle x - z, y - z\rangle = 0$, and

$$\|x - y\|^2 = \|x - z\|^2 + \|z - y\|^2, \qquad y \in P_F,$$

which proves the first part (with equality iff $y = z$). Also $\|z\|^2 = \sum_{\alpha\in F}|\langle x, u_\alpha\rangle|^2$, so putting $y = 0$ gives $\|x\|^2 = \|x - z\|^2 + \|z\|^2 \ge \|z\|^2$, the second inequality. ■

Inequality (14) says that the coefficients $c_\alpha$ making $\sum_\alpha c_\alpha u_\alpha$ closest to $x$ are $c_\alpha = \langle x, u_\alpha\rangle$. Since the second inequality holds for every finite $F \subset A$, Fact B.28 gives:

**Corollary 6.3.5 (Bessel's inequality)**

For any orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space $H$,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \le \|x\|^2, \qquad x \in H.$$

When is Bessel's inequality an equality? In finite dimensions, for a basis; here is the general analogue.

**Theorem 6.3.6**

For an orthonormal set $\mathcal{U} = \{u_\alpha : \alpha \in A\}$ in a Hilbert space $H$ the following are equivalent:

(a) for every $x \in H$,
$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|x\|^2; \tag{15}$$

(b) for all $x, y \in H$,
$$\sum_{\alpha\in A} \langle x, u_\alpha\rangle\overline{\langle y, u_\alpha\rangle} = \langle x, y\rangle; \tag{16}$$

(c) if $\mathcal{W} \supset \mathcal{U}$ is an orthonormal set, then $\mathcal{W} = \mathcal{U}$ (maximality);

(d) $P_A$ is dense in $H$.

**Proof.** For each $x \in H$ define $\widehat{x} : A \to \mathbb{C}$ by

$$\widehat{x}(\alpha) = \langle x, u_\alpha\rangle, \qquad \alpha \in A.$$

By Bessel's inequality $\widehat{x} \in \ell^2(A)$, and we get a bounded linear map

$$\Phi : x \mapsto \widehat{x} : H \to \ell^2(A). \tag{17}$$

(a) says $\Phi$ preserves norms; such a map preserves inner products by the polarization identity (10). The left side of (16) is exactly $\langle \widehat{x}, \widehat{y}\rangle$, so (a)⇒(b).

(b)⇒(c): if $\mathcal{W} \supsetneq \mathcal{U}$ is orthonormal, pick $u \in \mathcal{W} \setminus \mathcal{U}$; then (16) with $x = y = u$ reads $0 = \sum_\alpha |\langle u, u_\alpha\rangle|^2 = \|u\|^2 = 1$, a contradiction.

(c)⇒(d): if $P_A$ is not dense in $H$, Corollary 6.3.2 gives a unit vector $u \in (P_A)^\perp = (\overline{P_A})^\perp$, and $\mathcal{U} \cup \{u\}$ is a larger orthonormal set, so (c) fails.

(d)⇒(a): (15) holds on the dense subspace $P_A$ of finite combinations, so $\Phi$ is an isometry there. For $x \in H$ take $x_n \in P_A$, $x_n \to x$; as $\Phi$ is bounded,

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 = \|\widehat{x}\|_2^2 = \lim_{n\to\infty} \|\widehat{x_n}\|_2^2 = \lim_{n\to\infty} \|x_n\|^2 = \|x\|^2.$$

■

(16) is **Parseval's identity**, and an orthonormal set as in Theorem 6.3.6 is an **orthonormal basis**; then $\Phi$ in (17) preserves inner products, a **Hilbert space isomorphism** if onto. It is onto: for $y \in \ell^2(A)$ take finitely supported $y_n \to y$, $y_n = \Phi(x_n)$, $x_n \in P_A$; $\|x_n - x_m\| = \|y_n - y_m\|_2$, so $x_n \to x$ and $\Phi(x) = \lim_n y_n = y$. Hence:

**Theorem 6.3.7**

If $\{u_\alpha : \alpha \in A\}$ is an orthonormal basis of a Hilbert space $H$, then $x \mapsto \widehat{x}$ is a bijective Hilbert space isomorphism $H \to \ell^2(A)$.

So a Hilbert space is determined by the cardinality of an orthonormal basis; which basis is convenient depends on the situation.

### 6.3.4 Fourier series in $L^2(\mathbb{T})$

$\{u_n : n \in \mathbb{Z}\}$ (§5.1 (1)) is orthonormal in $L^2(\mathbb{T})$ with $\widehat{f}(n) = \langle f, u_n\rangle$, and Corollary 5.1.13 is Theorem 6.3.6(d): an orthonormal basis of $L^2(\mathbb{T})$. Explicitly:

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n)\overline{\widehat{g}(n)} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\overline{g(t)}\,dt, \qquad f, g \in L^2(\mathbb{T}),$$

also called **Parseval's identity** (the series converges absolutely for $f, g \in L^2(\mathbb{T})$, so the limit is unambiguous). Moreover

$$\lim_{n\to\infty} \Big\| f - \sum_{k=-n}^{n} \widehat{f}(k)u_k \Big\|_2 = 0, \qquad f \in L^2(\mathbb{T}). \tag{18}$$

(The partial sums converge in $\|\cdot\|_2$, so by Fact B.18 a subsequence converges a.e. In fact the Fourier series of an $L^2$ function converges a.e. (Carleson), while some $L^1$ function has an everywhere divergent Fourier series (Kolmogorov).)

Completeness played no big role in Theorem 6.3.6, but Theorem 6.3.7 for Fourier series gives, for every $a \in \ell^2(\mathbb{Z})$, an $L^2$ function with coefficients $a$; here completeness of $L^2(\mathbb{T})$ is decisive. With Fact B.17 this is the **Riesz–Fischer theorem**, unavailable for improper Riemann integrals: a main reason the Lebesgue integral is more useful.

### 6.3.5 The Plancherel transform

As $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ (Fact B.29), restricting gives the isomorphism $f \mapsto \widehat{f}$ and Parseval (18). As $L^2(\mathbb{R}) \not\subset L^1(\mathbb{R})$, on $\mathbb{R}$ we start from $C_c(\mathbb{R}) \subset L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

If $f \in C_c(\mathbb{R})$ and $\operatorname{supp} f \subset (-\pi, \pi)$, then Parseval for periodic functions and (19) of §5.2 (with $F = f$ on $(-\pi, \pi)$, $\widehat{F}(n) = \frac{1}{2\pi}\widehat{f}(n)$) give

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)|^2\,dt = \sum_{n=-\infty}^{\infty} \Big|\frac{1}{2\pi}\widehat{f}(n)\Big|^2.$$

This also holds for $g(x) = f(x)e^{-i\alpha x}$ (with $\widehat{g}(n) = \widehat{f}(n+\alpha)$), so

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2.$$

Integrating both sides over $\alpha \in [0, 1)$,

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \frac{1}{2\pi}\int_0^1 \sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\,d\alpha.$$

In general, if $\operatorname{supp} f \subset (-\pi\lambda, \pi\lambda)$, put $g(x) = \sqrt{\lambda} f(\lambda x)$; then $\operatorname{supp} g \subset (-\pi, \pi)$ and $\widehat{g}(\alpha) = \frac{1}{\sqrt{\lambda}}\widehat{f}(\alpha/\lambda)$, so for every $f \in C_c(\mathbb{R})$

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \int_{\mathbb{R}} |g(x)|^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{g}(\alpha)|^2\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\,d\alpha.$$

Let $\widehat{\mathbb{R}}$ denote the real line with $\frac{1}{2\pi}$ times Lebesgue measure. Then

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \int_{\widehat{\mathbb{R}}} |\widehat{f}(\alpha)|^2\,d\alpha, \qquad f \in C_c(\mathbb{R}), \tag{19}$$

and the inversion formula reads

$$f(x) = \int_{\widehat{\mathbb{R}}} \widehat{f}(\alpha) e^{i\alpha x}\,d\alpha.$$

Regarding $C_c(\mathbb{R})$ as a subspace of $L^2(\mathbb{R})$, (19) gives an isometric linear map

$$f \mapsto \widehat{f} : C_c(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, this map extends to an isometric linear map

$$\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

**Consistency on $L^1 \cap L^2$.** For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, is $\widehat{f} = \mathcal{F}(f)$? By the proofs of Facts B.20–B.22, for $p \neq q \in [1, \infty)$ and $f \in L^p[a, b] \cap L^q[a, b]$ there is a simple $s : [a, b] \to \mathbb{C}$ with simultaneously

$$\|f - s\|_p < \epsilon, \qquad \|f - s\|_q < \epsilon.$$

Moreover, choosing in the proof of Fact B.21 a continuous $g$ with

$$\mu(\{x : g(x) \neq s(x)\}) < \min\Big\{\Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^p, \Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^q\Big\}, \qquad |g| \leq \|s\|_\infty,$$

we get simultaneously

$$\|f - g\|_p < 2\epsilon, \qquad \|f - g\|_q < 2\epsilon. \tag{20}$$

Likewise (Fact B.22) every $f \in L^p(\mathbb{R}) \cap L^q(\mathbb{R})$ has $g \in C_c(\mathbb{R})$ with (20), so for $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ there are $g_n \in C_c(\mathbb{R})$ with

$$\|f - g_n\|_1 \to 0, \qquad \|f - g_n\|_2 \to 0,$$

and then

$$\lim_{n\to\infty} \|\widehat{f} - \widehat{g_n}\|_\infty = \lim_{n\to\infty} \|\mathcal{F}(f) - \mathcal{F}(g_n)\|_2 = 0.$$

Since $\widehat{g_n} = \mathcal{F}(g_n)$, we conclude $\widehat{f} = \mathcal{F}(f)$ (Exercise 6.3.4).

**Exercise 6.3.4**

If a sequence $\langle h_n \rangle$ in $C_0(\mathbb{R})$ satisfies $\lim_n \|f - h_n\|_\infty = \lim_n \|g - h_n\|_2 = 0$, show that $f = g$ almost everywhere.

**Solution** [Supplement]. Since $\|g - h_n\|_2 \to 0$, by Fact B.18 a subsequence $h_{n_k} \to g$ almost everywhere. But $h_{n_k} \to f$ uniformly, in particular pointwise everywhere. Hence $f = g$ almost everywhere. □

**Surjectivity.** By the proof of Theorem 5.2.8, every $h \in C_c^2(\mathbb{R})$ is $\widehat{k}$ for a bounded integrable, hence $L^2$, $k = \frac{1}{2\pi}\widehat{h}(-\cdot)$, so $h = \mathcal{F}(k)$. The range is dense in $L^2(\widehat{\mathbb{R}})$ and closed (isometric image of a complete space), so $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ is a surjective isometry (a Hilbert space isomorphism, by polarization), the **Plancherel transform**. Summarizing:

**Theorem 6.3.8 (Plancherel)**

There is a unique Hilbert space isomorphism $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ with $\mathcal{F}(f) = \widehat{f}$ for every $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

(Uniqueness: $\mathcal{F}$ is determined on the dense subspace $C_c(\mathbb{R}) \subset L^1 \cap L^2$ and is continuous.)

If $f \in L^2(\mathbb{R})$, then $f_A = \chi_{[-A,A]} f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and

$$\widehat{f_A}(\alpha) = \int_{-A}^{A} f(x) e^{-i\alpha x}\, dx.$$

Since $\lim_{A\to\infty} \|f - f_A\|_2 = 0$,

$$\lim_{A\to\infty} \|\mathcal{F}(f) - \widehat{f_A}\|_2 = 0,$$

the analogue of (18) for Fourier series. Finally, the statement that the Plancherel transform is a Hilbert space isomorphism reads

$$\int_{\mathbb{R}} f(x)\overline{g(x)}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}f(\alpha)\overline{\mathcal{F}g(\alpha)}\, d\alpha,$$

also called **Parseval's identity**.

**Example 2**

Applying Parseval to $f(x) = g(x) = e^{-|x|}$ (with $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$),

$$\int_{-\infty}^{\infty} (e^{-|x|})^2\, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{2}{1+\alpha^2}\right]^2 d\alpha,$$

so

$$\int_{-\infty}^{\infty} \frac{1}{(1+\alpha^2)^2}\, d\alpha = \pi \int_0^{\infty} e^{-2x}\, dx = \frac{\pi}{2}.$$

**Exercise 6.3.5**

Compute $\displaystyle\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha$ for $a, b > 0$.

**Solution** [Supplement]. By Example 1 of §5.2, $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$, real-valued. Parseval with $f = \chi_{[-a,a]}$, $g = \chi_{[-b,b]}$ (both in $L^1 \cap L^2$):

$$\int_{\mathbb{R}} \chi_{[-a,a]}\chi_{[-b,b]}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2\sin a\alpha}{\alpha} \cdot \frac{2\sin b\alpha}{\alpha}\, d\alpha.$$

The left side is $2\min\{a,b\}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2} \, d\alpha = \pi \min\{a,b\}.$$

(For $a = b$ this gives the classical $\int_{\mathbb{R}} \frac{\sin^2 a\alpha}{\alpha^2} \, d\alpha = \pi a$.) □